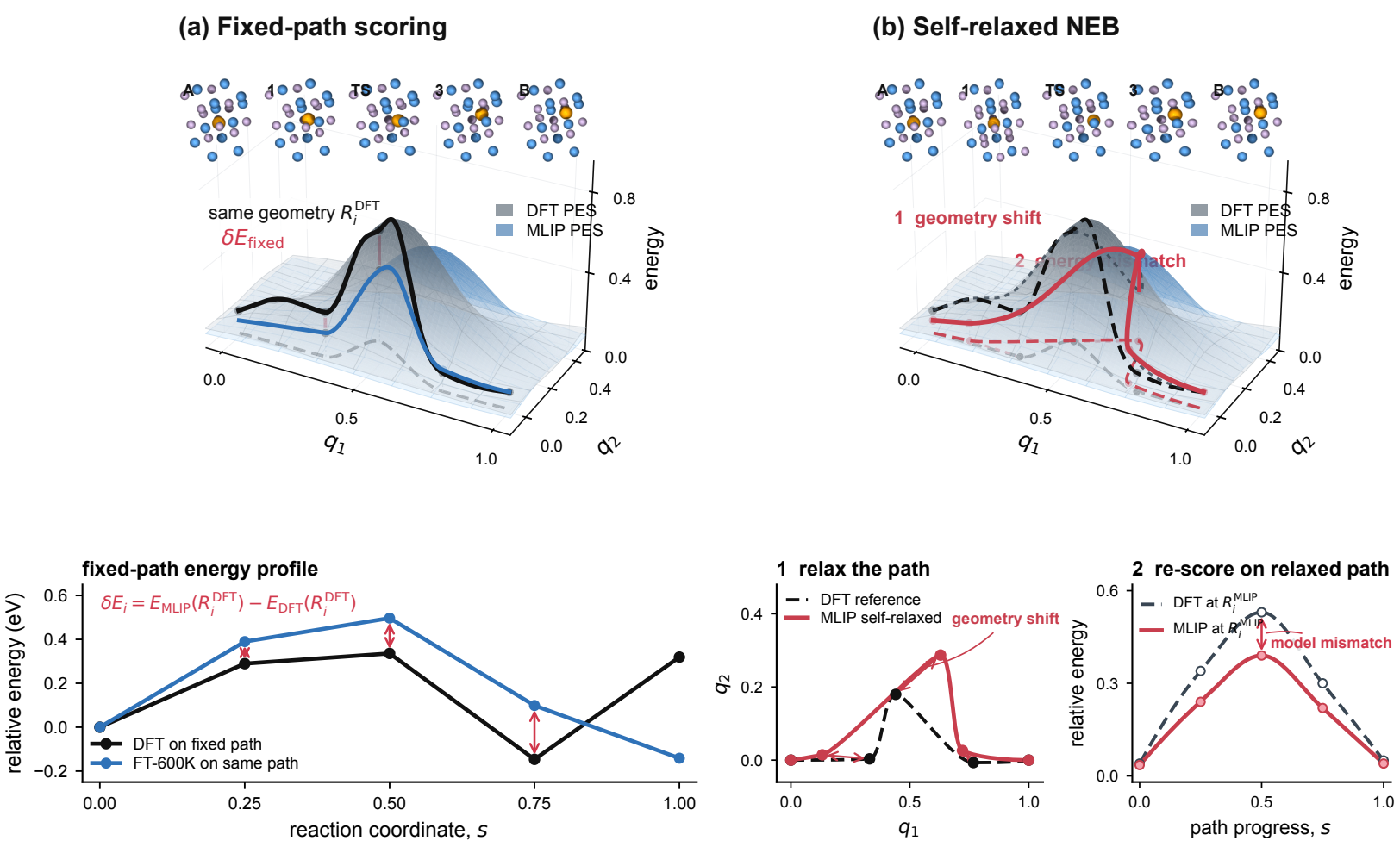

(a) Fixed-path scoring
A
1
TS
3
B
same geometry R_i^{DFT}
δE_{fixed}
DFT PES
MLIP PES
energy
0.8
0.4
0.0
0.0
0.5
1.0
q_1
0.4
0.2
0.0
q_2
(b) Self-relaxed NEB
1 geometry shift
2 energy mismatch
fixed-path energy profile
$\delta E_i = E_{MLIP}(R_i^{DFT}) - E_{DFT}(R_i^{DFT})$
relative energy (eV)
0.6
0.4
0.2
0.0
−0.2
DFT on fixed path
FT-600K on same path
0.00
0.25
0.50
0.75
1.00
reaction coordinate, s
1 relax the path
DFT reference
MLIP self-relaxed
geometry shift
q_2
q_1
2 re-score on relaxed path
DFT at R_i^{MLIP}
MLIP at R_i^{MLIP}
model mismatch
relative energy
path progress, s
$\Delta E_i = [E_{DFT}(R_i^{MLIP}) - E_{DFT}(R_i^{DFT})]_{geometry/path} + [E_{MLIP}(R_i^{MLIP}) - E_{DFT}(R_i^{MLIP})]_{energy\ model}$